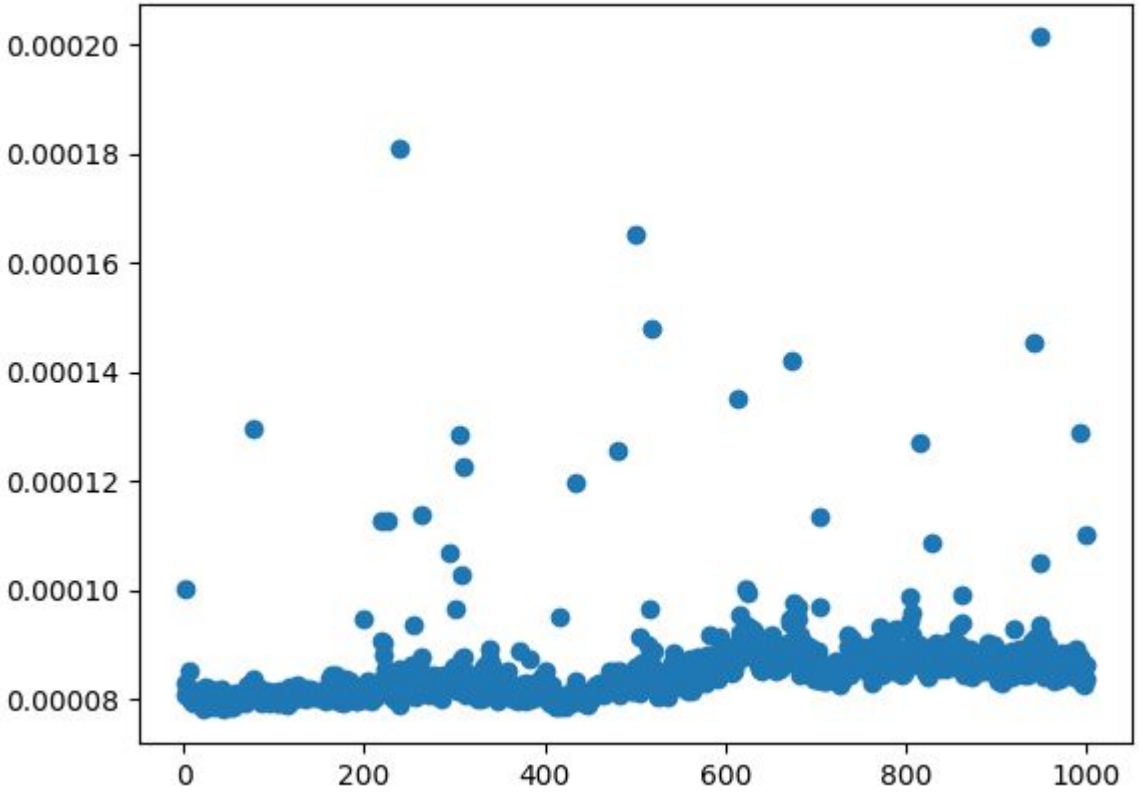
# Time-Complexity of Key-Insertion in Python Dicts

Rob Orsini

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ADVANCED LIGHT SOURCE

# Data summary from five runs (1k more per)

```
(lbl_tcoki) √ (roborsini@Robs-Mac-mini.local:~/projects/lbl_tcoki)
% ./tcoki.py                                                                    [main] lbl_tcoki
{'dict size': [1, 1001], 'std(time)': np.float64(7.218238577676717e-06), 'mean': array([5.00500000e+02, 8.55799295e-05])}
{'dict size': [1001, 2001], 'std(time)': np.float64(9.572801763083436e-06), 'mean': array([1.50050000e+03, 8.83417011e-05])}
{'dict size': [2001, 3001], 'std(time)': np.float64(9.155585199761651e-06), 'mean': array([2.50050000e+03, 9.09389696e-05])}
{'dict size': [3001, 4001], 'std(time)': np.float64(1.7625068531240522e-05), 'mean': array([3.50050000e+03, 9.41751027e-05])}
{'dict size': [4001, 5001], 'std(time)': np.float64(9.352592786659284e-06), 'mean': array([4.50050000e+03, 9.22965202e-05])}
(lbl_tcoki) √ (roborsini@Robs-Mac-mini.local:~/projects/lbl_tcoki)
%                                                                              [main] lbl_tcoki
```

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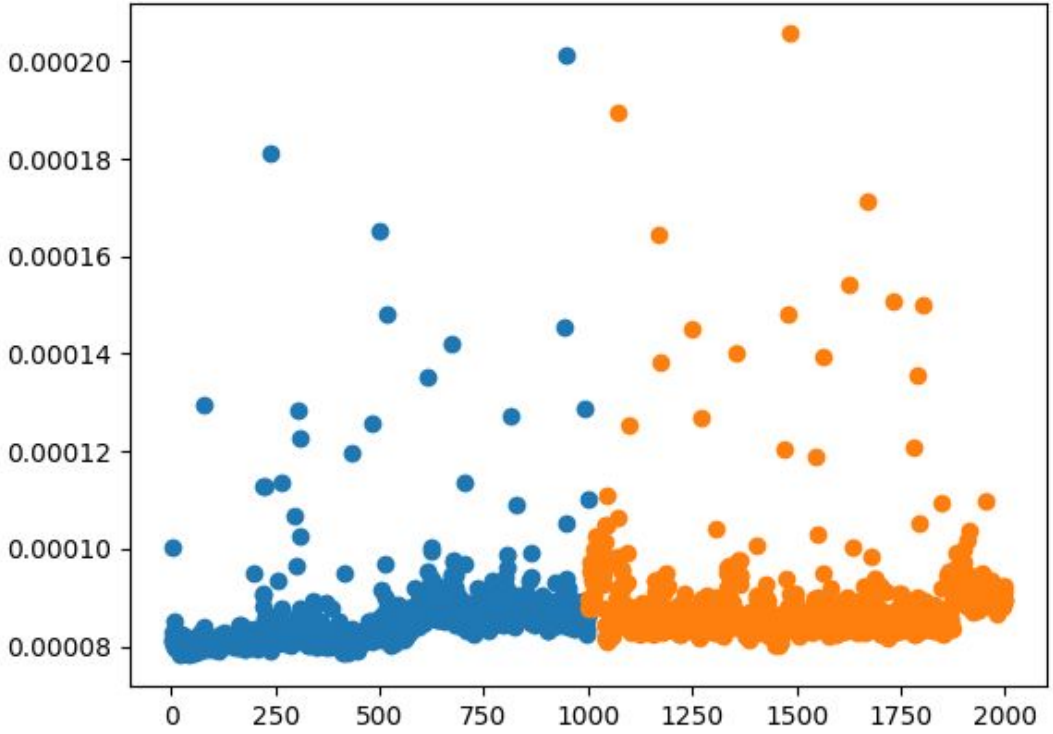


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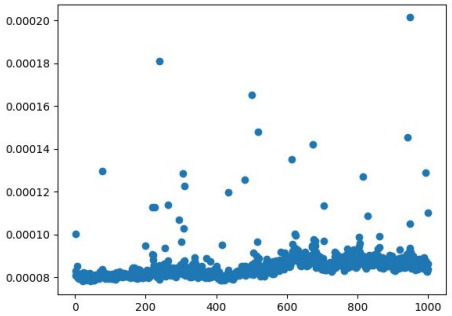


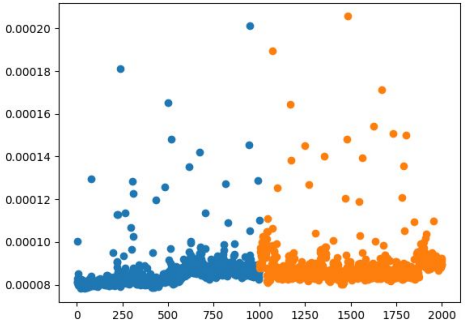


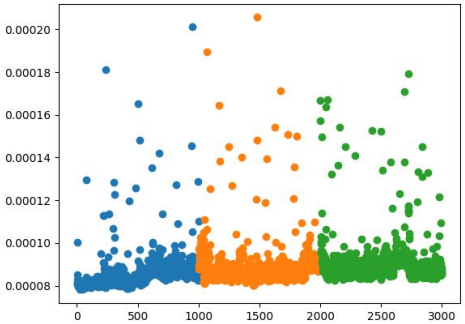


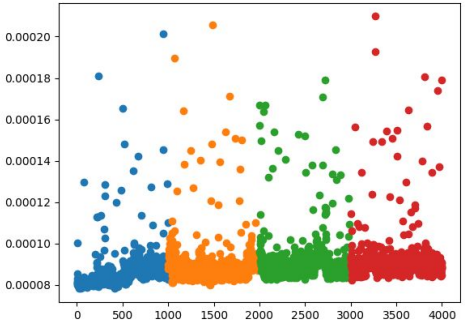


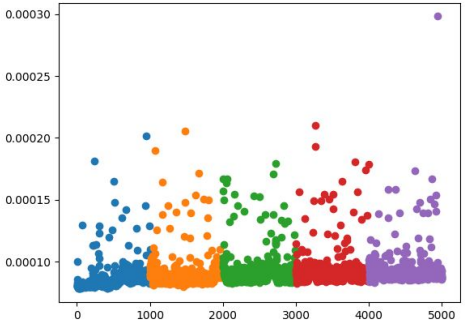


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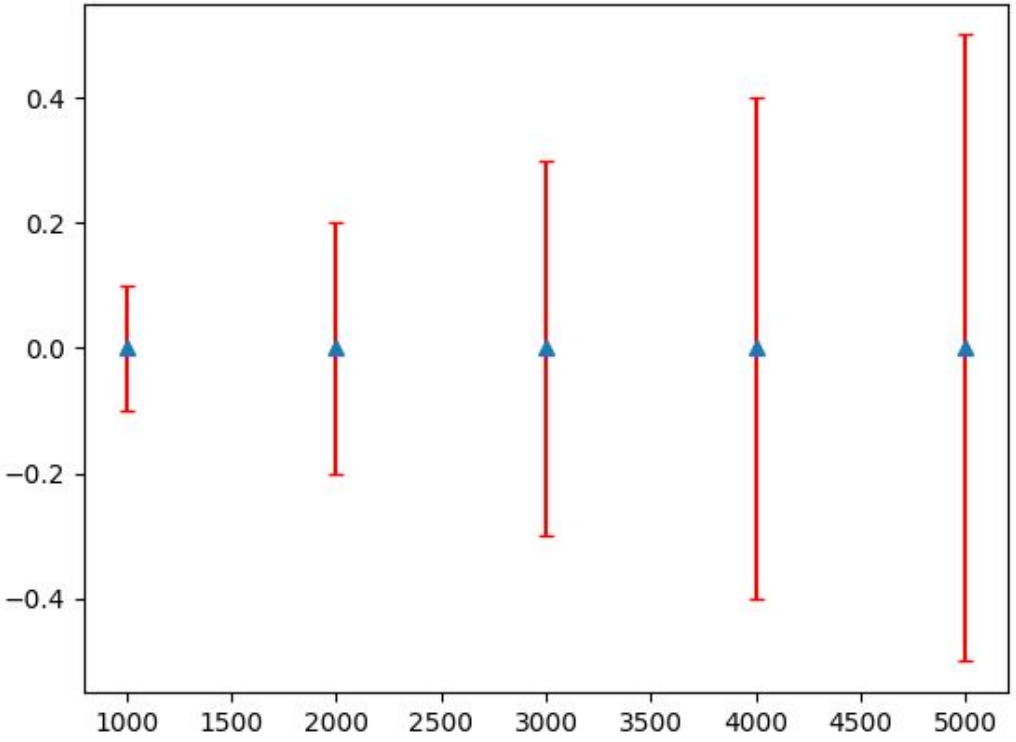
# A plot of mean and standard deviation per bucket

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## The time-complexity of key-insertion in Python dictionaries is constant time, or..

# $O(1)$

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